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Shammo Dutta

EDUCATION

The University of Alabama (UA)

Doctor of Philosophy in Mechanical Engineering

GPA: 3.93/4.0

Lab: Smart Materials and Robotic Technologies (SMART) Lab

Advisor: Dr. Sree Kalyan Patiballa

Tuscaloosa, U.S.

Aug. 2022-Jul. 2026 (Expected)

The University of Alabama (UA)

Master of Science in Mechanical Engineering

GPA: 3.93/4.0

Tuscaloosa, U.S.

Awarded Dec. 2025

Amrita Vishwa Vidyapeetham (University)

Bachelor of Technology in Mechanical Engineering

GPA: 9.0/10.0

Amritapuri, India

Jul. 2017-Jun. 2021

RESEARCH INTERESTS

Mechanical metamaterials · Hierarchical metamaterials · Inverse design for nonlinear response and shape matching · Soft robotics modeling and fabrication · Design optimization

RESEARCH EXPERIENCE

SMART Lab, UA

Graduate Research Assistant | Advisor: Dr. Sree Kalyan Patiballa

Tuscaloosa, U.S.

Aug. 2022-Present

- Developing a homogenization and qualitative library-driven design framework for 3D hierarchical mechanical metamaterials to achieve target mechanical responses.
- Developed a high-fidelity finite element (FEA) framework for tendon-driven soft actuators.
- Modeled and analyzed dynamic impact behavior of civil metamaterial-based sandwich composites.
- Devised a robust stress-based design optimization method for planar mechanical metamaterials.

Department of Mechanical Engineering, Amrita University

Research Assistant | Advisor: Dr. Hariprasad M.P.

Amritapuri, India

Sep. 2021-Jul. 2022

- Co-authored a technical guide on computational tools for experimental mechanics, emphasizing 2D digital image correlation (DIC) for full-field deformation measurement.
- Collaborated in interdisciplinary projects analyzing load flow and stress concentration in implant-supported structures using DIC and FEA.

Department of Mechanical Engineering, Amrita University

Undergraduate Researcher | Advisor: Dr. Hariprasad M.P.

Amritapuri, India

Aug. 2019-Jun. 2021

- Developed lightweight load-bearing structures reinforced with strut-based auxetic metamaterials.
- Co-invented one U.S. patent, Indian patent and, communicated a high-impact journal article.

PUBLICATIONS

JOURNALS

- [J1] **Dutta, S.**, Krishnan Girish, Patiballa, S. K. (2026). A Qualitative library based Design Optimization Framework for Hierarchical Mechanical Metamaterials. Manuscript in-preparation, Planned for *ASME Journal of Mechanical Design*
- [J2] Turna, N. R., Luong, T., **Dutta, S.**, Aaleti, S., & Patiballa, S. K. (2026). Auxetic Metamaterials Embedded Concrete Systems (AuxMeCs) for Resilient Civil Infrastructures. Manuscript in-preparation, Planned for *Wiley Advanced Engineering Materials*.
- [J3] **Dutta, S.**, Conzola, J., Vikas, V., & Patiballa, S. K. (2025). Development and Validation of Computational Modeling Framework for Tendon-based Soft Robotics. *ASME Journal of Mechanisms and Robotics*. <https://doi.org/10.1115/1.4069292>
- [J4] **Dutta, S.**, & Patiballa, S. K. (2025). Design Optimization for Uniform Stress Distribution in Mechanical Metamaterials Using a Maximum Material Utilization Metric. *ASME Journal of Mechanical Design*, 148(1): 011702. <https://doi.org/10.1115/1.4068956>
- [J5] Baghiana G, Manju V, Hariprasad M.P., Menon H.G., **Dutta S**, Vinod Kumar Gopal (2022) Comparison of Attachment Types in Maxillary Implant-assisted Obturators using Digital Image Correlation Analysis. *J Contemp Dent Pract*, 23(7), 695-702. <https://pubmed.ncbi.nlm.nih.gov/36440515/>
- [J6] Menon, H. G., **Dutta, S.**, Krishnan, A., M. P., H., Shankar, B. (2022). Proposed auxetic cluster designs for lightweight structural beams with improved load bearing capacity. *Engineering Structures*, 260, 114241. <https://doi.org/10.1016/j.engstruct.2022.114241>

CONFERENCE PROCEEDINGS

- [C1] **Dutta, S.**, & Patiballa, S. K. (2025). Towards self-sensing in auxetic metamaterials using displacement mapping framework. Volume 5: 21st IEEE/ASME International Conference on Mechatronic and Embedded Systems and Applications (MESA); 49th Mechanisms and Robotics Conference (MR). <https://doi.org/10.1115/DETC2025-169766>
- [C2] **Dutta, S.**, & Patiballa, S. K. (2023). Design optimization framework for uniform stress distribution of mechanical Metamaterials. Volume 4: Advanced Materials: Design, Processing, Characterization and Applications; Advances in Aerospace Technology. <https://doi.org/10.1115/imece2023-112793>
- [C3] Menon H.G., **Dutta S**, Hariprasad M.P., Shankar B. (2021). Investigation on deflection characteristics of auxetic beam structures using FEM. Recent Advances in Manufacturing, Automation, Design and Energy Technologies, 621-628. https://doi.org/10.1007/978-981-16-4222-7_70
- [C4] **Dutta, S.**, Menon, H. G., Hariprasad, M., Krishnan, A., Shankar, B. (2021). Study of auxetic beams under bending: A finite element approach. *Materials Today: Proceedings*, 46, 9782-9787. <https://doi.org/10.1016/j.matpr.2020.10.479>

PATENTS

- [P1] **Dutta, S.**, MP, H., Shankar, B., Menon, H. G. (2025). Auxetic Member for Load Bearing Structures. IN. Patent No. 575791. Patent Office, Government of India. (Status: Granted, Filed in 2021)
- [P2] Shankar, B., MP, H., **Dutta, S.**, Menon, H. G. (2024). Auxetic Member for Load Bearing Structures. U.S. Patent No. 11981111. U.S. Patent and Trademark Office. (Status: Granted, Filed in 2021)
<https://patents.google.com/patent/US11981111B2/en>

CONFERENCE PRESENTATIONS

1. ASME 2025 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC-CIE) *Anaheim, California* [Aug. 17-20, 2025]
2. Embodied Intelligence Conference (EI) 2025 by Bio-Inspired Robotics Laboratory, University of Cambridge, *Held in Cambridge U.K., Attended virtually* [Apr. 02-04, 2025]
3. ASME 2024 Smart Materials, Adaptive Structures and Intelligent Systems (SMASIS), *Atlanta, Georgia* [Sep. 9-11, 2024]
4. ASME 2023 International Mechanical Engineering Congress & Exposition (IMECE), *New Orleans, Louisiana* [Oct. 29, 2023-Nov. 2, 2023]
5. ASME 2022 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC-CIE), *St. Louis, Missouri* [Aug. 14-17, 2022]
6. International Conference on Future Technologies in Manufacturing, Automation, Design & Energy, *National Institute of Technology Puducherry* [Dec. 28-30, 2020]
7. International Mechanical Engineering Congress 2019, *National Institute of Technology Tiruchirappalli* [Nov. 29, 2019-Dec. 1, 2019]

TEACHING EXPERIENCE

ME 489 & 490 - Mechanical Engineering Design I & II

Graduate Teaching Assistant, Department of Mechanical Engineering, UA

Aug. 2025-Present

- Mentoring 50+ senior engineering students through capstone design projects, evaluating technical feasibility, innovation, and presentation quality.

ME 372 - Dynamic Systems

Graduate Teaching Assistant, Department of Mechanical Engineering, UA

Aug. 2024-May. 2025

- Developed supplementary MATLAB tutorials for dynamic systems modeling, improving student performance by providing hands-on computational skills training.
- Evaluated 70+ assignments per term and provided detailed feedback to individual students.

ME 349 - Engineering Analysis

Graduate Teaching Assistant, Department of Mechanical Engineering, UA

Aug. 2023-Aug. 2024

- Graded 100+ assignments per term and provided detailed feedback to individual students.

ME 215 - Thermodynamics I

Graduate Teaching Assistant, Department of Mechanical Engineering, UA

Jun. 2023-Jul. 2023

- Evaluated over 50 assignments per term, including homework, exams, and final assessments.

FELLOWSHIP AND AWARDS

Graduate Council Conference and Research Fund

2025

UA Graduate School and Department of Mechanical Engineering for ASME IDETC 2025.

Graduate Council Conference and Research Fund

2024

UA Graduate School and Department of Mechanical Engineering for ASME SMASIS 2024.

Graduate Council Conference and Research Fund

2023

UA Graduate School and Department of Mechanical Engineering for ASME IMECE 2023.

Graduate Council Fellowship

2022-2023

Awarded by the UA Graduate School; merit-based fellowship for incoming doctoral students.

SKILLS

- **Computational Tools:** FEA (ABAQUS, COMSOL), MATLAB (Numerical methods, local and global optimization), Python (NumPy, SciPy, Matplotlib, CMA-ES, OpenCV), CAD (Solidworks, Autodesk Fusion), Unity/VR-based design
- **Fabrication & Experimental Techniques:** 3D printing (FDM, SLA), Silicone casting and soft lithography, Digital Image Correlation (DIC), ASTM standard material testing
- **Visualization & Post-Processing:** Inkscape, Origin, Paraview, Adobe Creative Cloud
- **Languages:** English, Hindi, Marathi (Professional); Bengali (Native); Malayalam (Basic)

PROFESSIONAL MEMBERSHIP

- American Society of Mechanical Engineers (ASME)
- U.S. Association for Computational Mechanics (USACM)

REFERENCES

1. Prof. Sree Kalyan Patiballa

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2. Prof. Vishesh Vikas

Associate Professor

Director, Agile Robotics Laboratory (ARL) Lab

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3. Prof. Sriram Aaleti

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